

Hands on Create authentication service that returns JWT

AuthenticationController.java

```
package com.cognizant.spring_learn.controller;

import io.jsonwebtoken.Jwts;
import io.jsonwebtoken.security.Keys;
import io.jsonwebtoken.security.MacAlgorithm;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;

import javax.crypto.SecretKey;
import java.nio.charset.StandardCharsets;
import java.util.Base64;
import java.util.Date;
import java.util.HashMap;
import java.util.Map;

@RestController
public class AuthenticationController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(AuthenticationController.class);

    private static final SecretKey SECRET_KEY =
        Keys.hmacShaKeyFor(
            "secretkeysecretkeysecretkey123456"
            .getBytes(StandardCharsets.UTF_8));

    @GetMapping("/authenticate")
    public Map<String, String> authenticate(
        @RequestHeader("Authorization") String authHeader) {

        LOGGER.debug("START");
        LOGGER.debug("Authorization Header : {}", authHeader);
```

```

String user = getUser(authHeader);

String token = generateJwt(user);

Map<String, String> map = new HashMap<>();
map.put("token", token);

LOGGER.debug("END");

return map;
}

private String getUser(String authHeader) {

    LOGGER.debug("START getUser()");

    String encoded = authHeader.substring("Basic ".length());

    byte[] decoded = Base64.getDecoder().decode(encoded);

    String credentials =
        new String(decoded, StandardCharsets.UTF_8);

    LOGGER.debug("Decoded Credentials : {}", credentials);

    String user = credentials.split(":")[0];

    LOGGER.debug("User : {}", user);
    LOGGER.debug("END getUser()");

    return user;
}

private String generateJwt(String user) {

    LOGGER.debug("START generateJwt()");

    String token = Jwts.builder()
        .subject(user)
        .issuedAt(new Date())
        .expiration(new Date(System.currentTimeMillis() + 1200000))

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        .signWith(SECRET_KEY, Jwts.SIG.HS256)
        .compact();

    LOGGER.debug("Generated Token : {}", token);
    LOGGER.debug("END generateJwt()");

    return token;
}
}

```

SecurityConfig.java

```

package com.cognizant.spring_learn.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.Customizer;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.core.userdetails.User;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.core.userdetails.UserDetailsService;
import org.springframework.security.provisioning.InMemoryUserDetailsManager;
import org.springframework.security.web.SecurityFilterChain;

@Configuration
public class SecurityConfig {

    @Bean
    public SecurityFilterChain securityFilterChain(HttpSecurity http) throws Exception {

        http
            .csrf(csrf -> csrf.disable())
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/authenticate").hasAnyRole("USER", "ADMIN")
                .anyRequest().authenticated())
            .httpBasic(Customizer.withDefaults());

        return http.build();
    }

    @Bean
    public UserDetailsService users() {

```

```

    UserDetails user = User.withUsername("user")
        .password("{noop}pwd")
        .roles("USER")
        .build();

    UserDetails admin = User.withUsername("admin")
        .password("{noop}pwd")
        .roles("ADMIN")
        .build();

    return new InMemoryUserDetailsManager(user, admin);
}
}

```

Pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>4.1.0</version>
        <relativePath/> <!-- lookup parent from repository -->
    </parent>
    <groupId>com.cognizant</groupId>
    <artifactId>spring-learn</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <name/>
    <description/>
    <url/>
    <licenses>
        <license/>
    </licenses>
    <developers>
        <developer/>
    </developers>
    <scm>

```

```
<connection/>
<developerConnection/>
<tag/>
<url/>
</scm>
<properties>
  <java.version>17</java.version>
</properties>
<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-webmvc</artifactId>
  </dependency>

  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-devtools</artifactId>
    <scope>runtime</scope>
    <optional>true</optional>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-webmvc-test</artifactId>
    <scope>test</scope>
  </dependency>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
  </dependency>
  <dependency>
    <groupId>io.jsonwebtoken</groupId>
    <artifactId>jjwt-api</artifactId>
    <version>0.12.6</version>
  </dependency>

  <dependency>
    <groupId>io.jsonwebtoken</groupId>
    <artifactId>jjwt-impl</artifactId>
    <version>0.12.6</version>
    <scope>runtime</scope>
  </dependency>
```

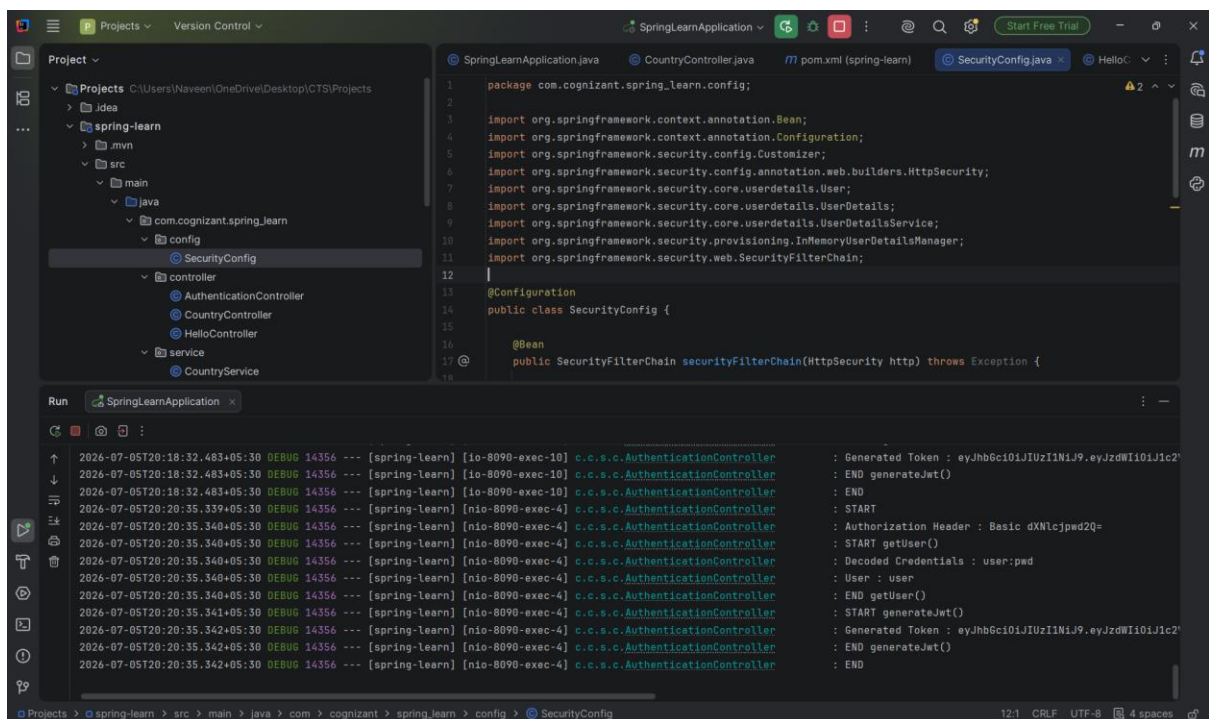
```

<dependency>
  <groupId>io.jsonwebtoken</groupId>
  <artifactId>jjwt-jackson</artifactId>
  <version>0.12.6</version>
  <scope>runtime</scope>
</dependency>
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-security</artifactId>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>

</project>

```



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Naveen>curl -u user:pwd http://localhost:8090/authenticate
{"token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiIj1c2VyIiwiaWF0IjoxNzgzMjYyOTYLC3leHAiOiJlE3ODMyNjQxMTJ9.4BnYfUFZrUSLp1H_H8F61cPsCjgGA7v2JMyfUhhSk0o"}
C:\Users\Naveen>
```

